Nazia Shehnaz Joynab

nsjoynab | nazia.joynab@utdallas.edu | geek-a-byte | +1 (814) 405-9890 | Portfolio

Education

- **University of Texas, Dallas** — Dallas, TX
  Master of Science in Computer Science — *Spring 2026 - Summer 2027 (Exp.)*
  - Graduate Researcher @ Dr. Latifur Khan's Lab. **(Summer'26)**
  - Courses Taken: Design and Analysis of Computer Algorithms, Artificial Intelligence, Machine Learning, Robotics
- **Military Institute of Science and Technology** — Dhaka, Bangladesh
  Bachelor of Science in Computer Science and Engineering; CGPA: 3.65/4.00 — *Feb. 2019 – May 2023*

Research Interests

• AI/ML • NLP • Security • Reinforcement Learning • Computer Vision • Robotics

Experience

- **Samsung R&D Institute Bangladesh** — Dhaka, Bangladesh

  ***Software Engineer I (DevOps)**, Full-time, Onsite* — *Dec 2023 - Oct 2025*
  - Contributed to the development and maintenance of an internal content delivery network named ADS (Artifact Deployment System) which leverages advanced caching methods, proxies and cloud technologies to optimize transmission of large binary artifacts across 23 global R&D centers of Samsung Electronics.
  - Engineered an admin portal and CI plugin using Django and J2EE, cutting cloud storage cost by ensuring consistent artifact retention across on-premise and cloud infrastructures (AWS S3).
  - Automated a complex multi-step artifact migration process between storage servers using Bash and Fabric, achieving a **1.75x** speed improvement compared to standard transfer methods.
  - Engaged in comprehensive system monitoring, shouldering critical communication with overseas stakeholders.
  - Actively participated in peer programming, code review, and discussions related to emergency issue mitigation in the production and efficient system design.

  ***Full Stack Development Intern**, Full-time, Onsite* — *Jun 2023 - Dec 2023*
  - Designed and implemented a secure and efficient patent and ideation file management tool using CodeIgniter and MySQL.

Publications

- Munir, M. B., Ononto, A. J., **Joynab, N. S.**, Thuraisingham, B., & Khan, L. (2026). TRIS: A Tri-Layer Retrieval Integrity Sieve Against Knowledge Poisoning. *Findings of the Association for Computational Linguistics: EMNLP 2026*. (link)
- **Joynab, N. S.**, Islam, M. N., Aliya, R. R., Hasan, A. R., Khan, N. I., & Sarker, I. H. (2024). A federated learning aided system for classifying cervical cancer using pap-smear images. *Informatics in Medicine Unlocked*. (link , pdf , data and materials)

Projects

- **Federated learning aided Cervical Cancer classification (Undergrad Thesis)**
  - Proposed a CNN-based federated learning architecture for cervical cancer classification which showed test accuracy of **94.36%** and **78.4%** on an IID (Independent and Identically Distributed) and a non-IID setting respectively.
- **TRIS: Tri-Layer Retrieval Integrity Sieve**
  - Contributed to TRIS, a tri-layer middleware defense for mitigating knowledge poisoning attacks in Retrieval-Augmented Generation (RAG) systems.
  - Implemented and evaluated retrieval integrity mechanisms using semantic clustering, structural trigger–payload filtering, and LLM consistency verification.
  - Conducted adversarial evaluations across NQ, HotpotQA, and MS-MARCO, achieving attack success rates as low as 3% on NQ and 4% on MS-MARCO in the evaluated black-box setting.

- **Joyita: An ML based screening application for facilitating the Cervical Cancer treatment**
  - Developed an android application with re-usable widgets in **Flutter**. Used **Cloud Firestore** to store doctor's appointment details, patient blogs etc. The ML model predicted the possibility of having cervical cancer based on user inputs of a pre-defined questionnaire displayed on the app's UI. Hosted ML model using **Flask and Heroku**.
  - Testing of the application was done by almost **50+** users.

Programming Skills

- **Programming Languages**: Python, Java, Bash, C++, SQL, JavaScript
- **ML Frameworks/Libraries**: Pandas, Scikit-learn, Keras, Tensorflow, Pytorch, LangChain, Agno
- **Robotics & Simulation:** ROS 2, NVIDIA Isaac Sim
- **Backend Frameworks**: Django, Laravel, CodeIgniter, Flask
- **Databases**: MySQL, Postgresql, ChromaDB, Oracle, Firebase
- **Android**: Dart, Flutter
- **DevOps**: Linux, Docker, Kubernetes (Basic), AWS S3 (Basic), Quickbuild, GitHub, Splunk

Achievements

- **Tech Innovator**, MX SE Award for ADS cloud mode development, Samsung Electronics (2025) [link]
- **2nd Runners-up**, Devmania by BDOSN (2022)
- **34th position**, Inter University Girls' Programming Contest Season 2, NSU (2019)

Problem Solving Profile

Leetcode , **Hackerrank** , **Vjudge** , **CSES** (Solved **600 problems** across all platforms.)